

Simulating an Ant Colony

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Theoretical Background / Motivation

- Ant colonies show emergent behaviour such as foraging without centralized control.
- Stigmergy: indirect coordination via environment modification.
 - Ants deposit pheromones during food trips.
 - Shorter paths accumulate stronger signals which are reinforced by more ants, enabling convergence.
- For our simulation, metrics like time to shortest path, average trip lengths, and efficiency (food per energy consumed) are used to evaluate the performance of the colony.
- These observables help capture emergent behaviour and compare different model extensions / variants.

Theoretical Background / Motivation

- Ant Colony behaviour is studied for: Evolutionary Biology, Emergent Phenomena, Robotics and Swarm Intelligence, Ant Colony Optimization (optimal path finding) etc.

Our Research Question / Motivation

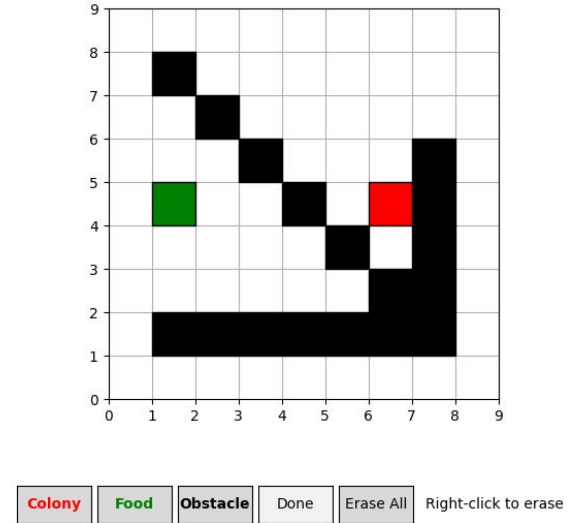
“How do the dynamics of the ant colony get influenced by varying different parameters?”

Disclaimer

There are MANY different ways to simulate ant colonies. Our results, while capture expected behaviour, may not directly correspond to those found in the literature.

Basic Simulation Setup

- Environment: Colony, Food, Obstacles
- Ant Behaviour:
 - Weighted Random Walk
 - When Food is picked:
 - Pheromone dropped (strength = $1/\text{length}$)
 - Travel back the same path
 - If no food found return back. The ant can die of hunger.
 - When an ant reaches the colony, it forgets the path it had travelled.
 - By default, the ants are not allowed to immediately step back.



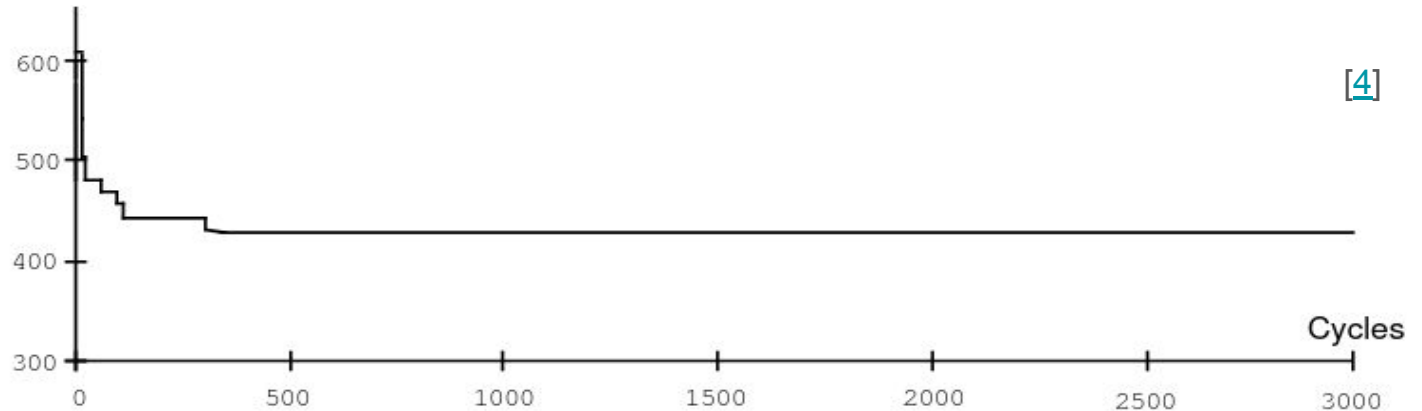
Observables Studied

- Ant Efficiency
 - Shortest Path over time
 - Average length of latest successful trips by all ants
-
- Amount of Food Collected by the Colony over time
 - Ratio of Successful Trips / Total Trips
 - Area Visited over Time
 - Ant Population

Computing errors

Expected Behaviour (Literature)

- Convergence to short path [1][3][4]
- Improvement from more ants [2]
- Inter-ant communication improves results [4]



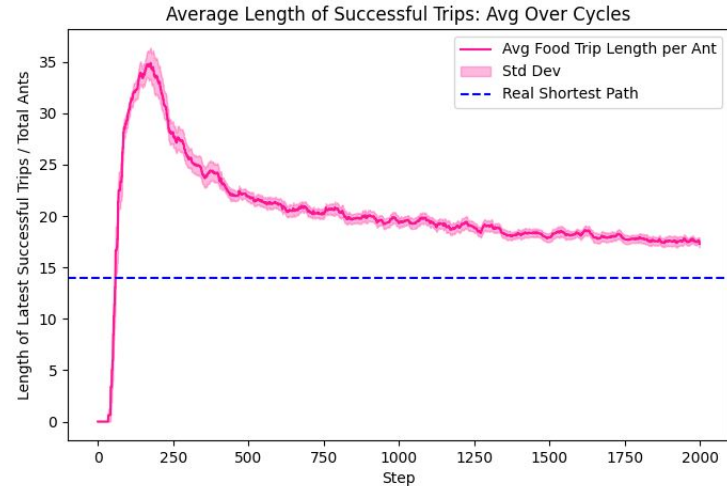
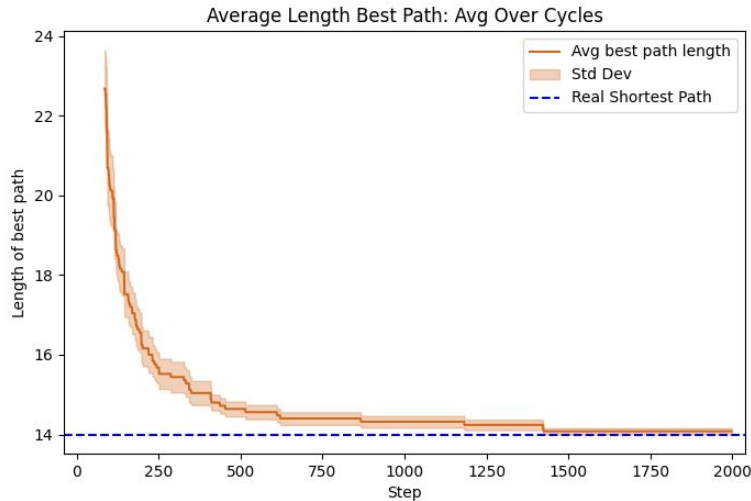
Sample Results

Real Shortest Path Length to Food: 14

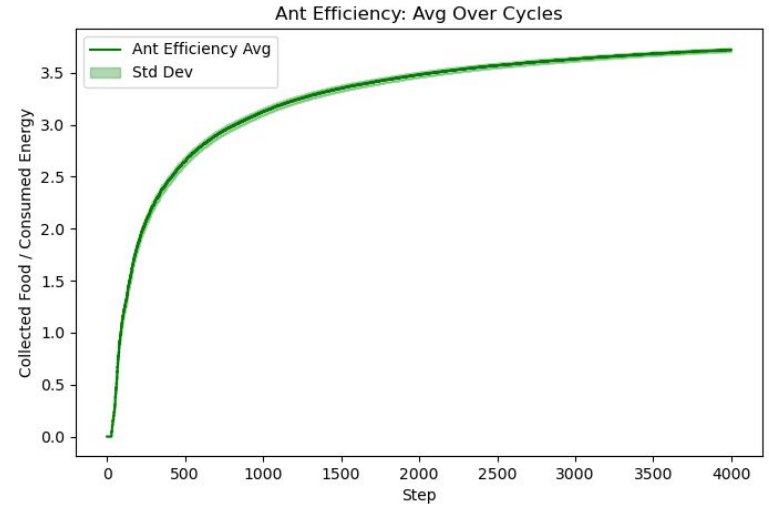
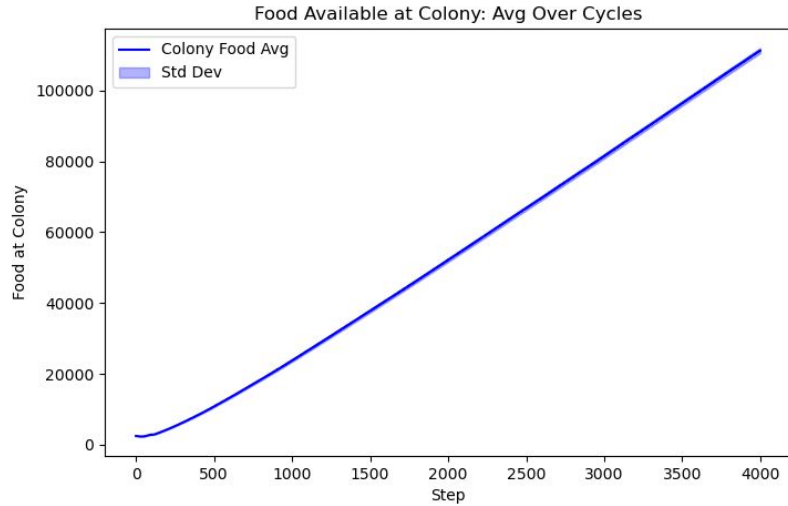
Average Shortest Path Length found by Ants: 14.08 +/- 0.08

Average Time to First Path: 57.72 +/- 2.49

Average Time to Shortest Path: 427.72 +/- 77.63

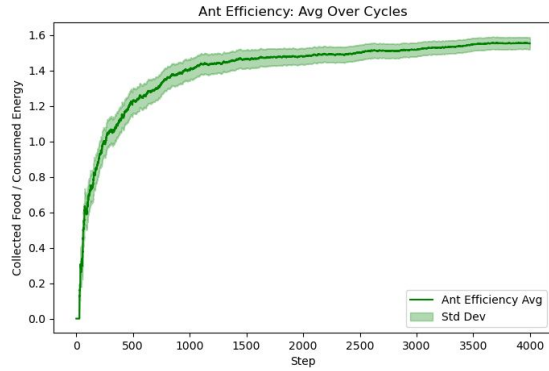


Sample Results

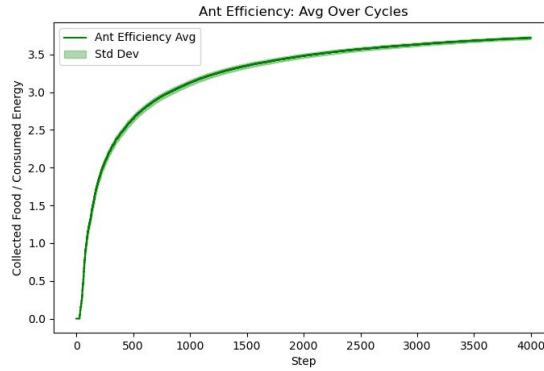


Results

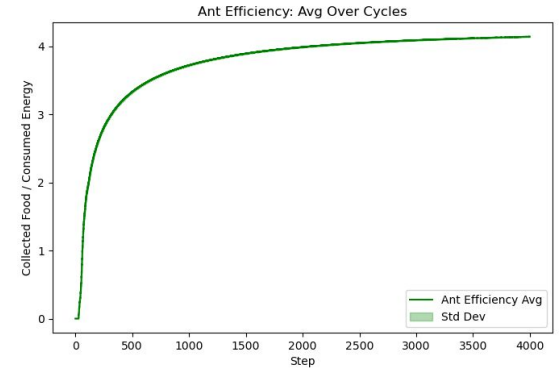
Number of Ants



1 Ant

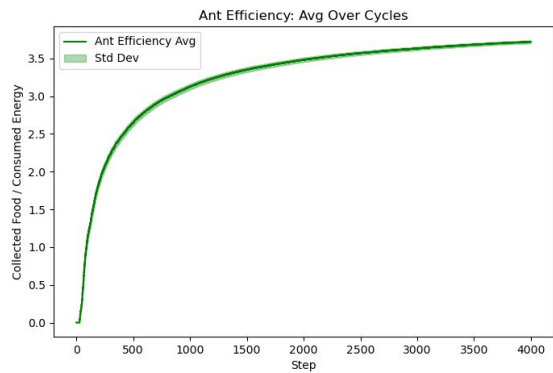


10 Ants

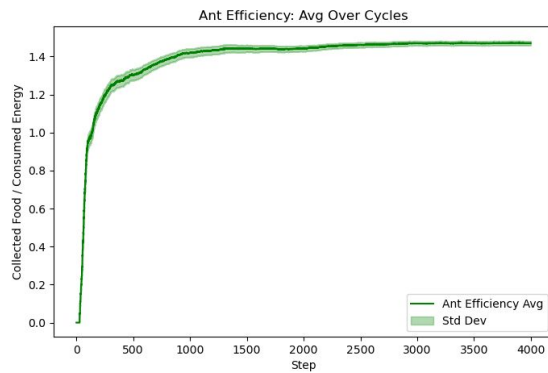


100 Ants

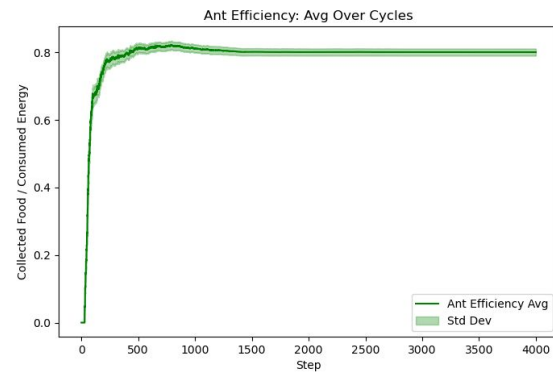
Pheromone Decay



0.01

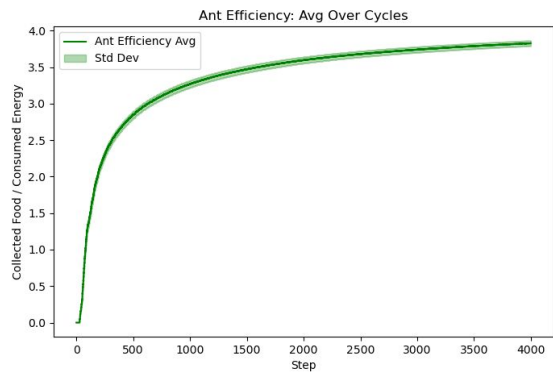


0.1

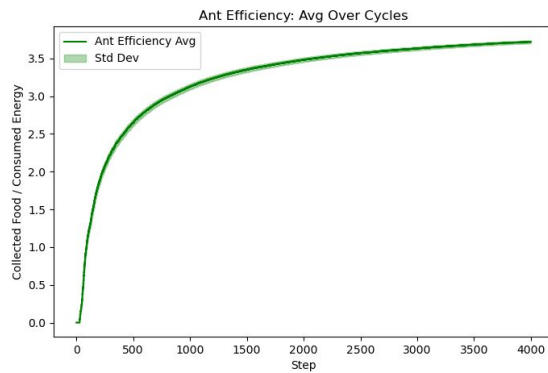


0.5

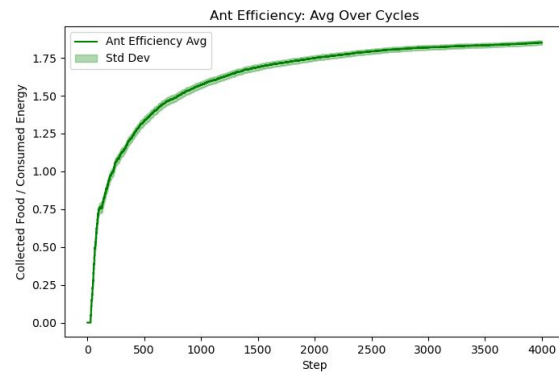
Exploration Desire



0.001

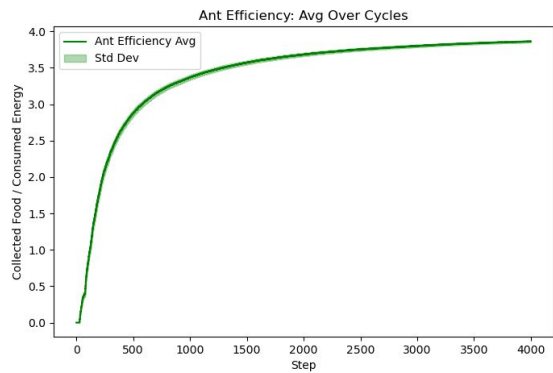


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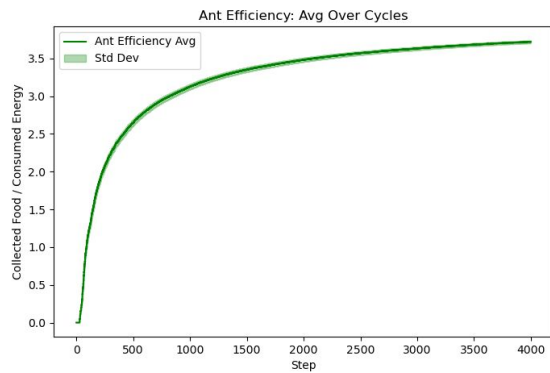


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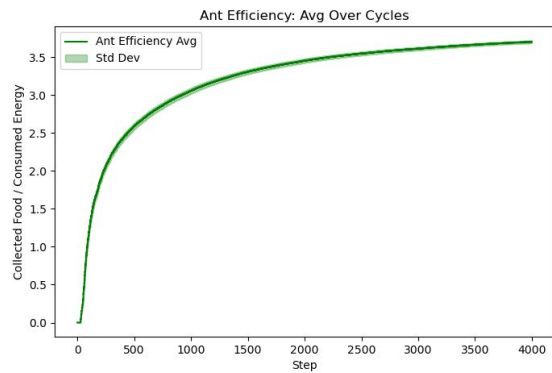
Ant Energy



50

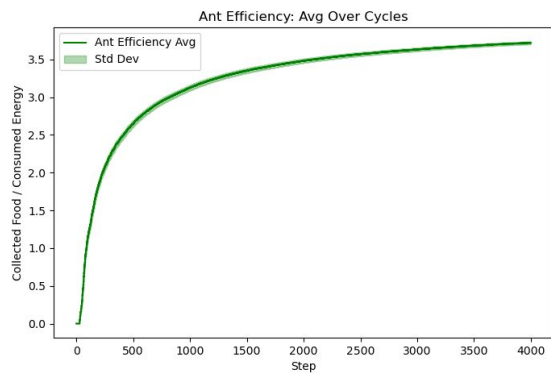


100

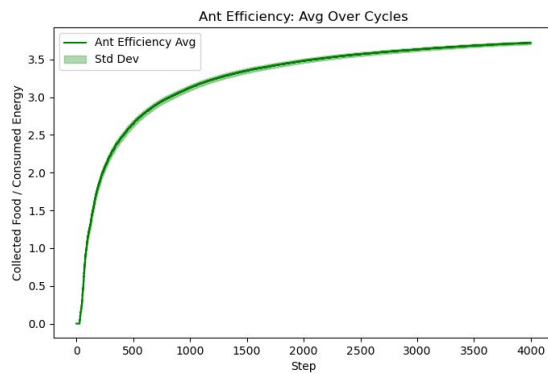


500

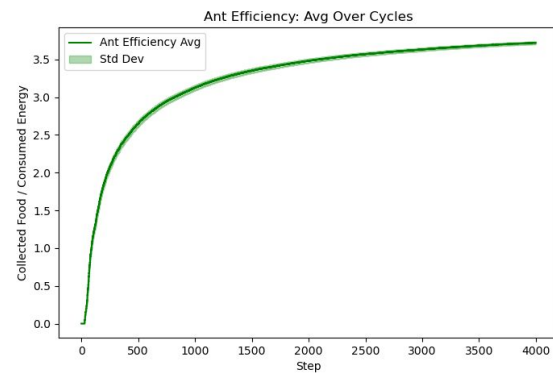
Colony Start Amount



100

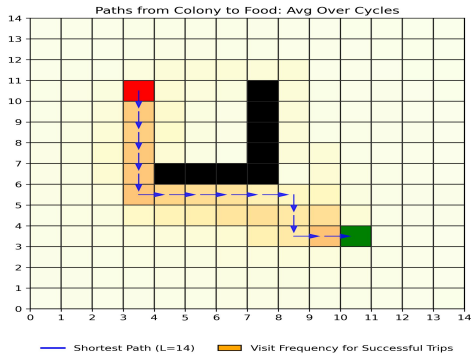
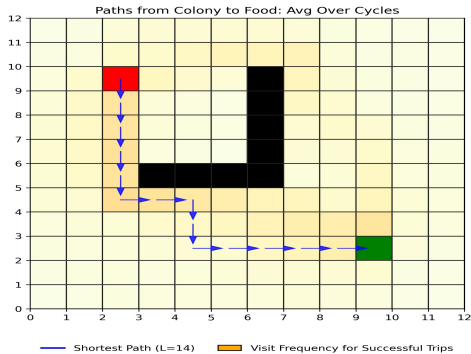
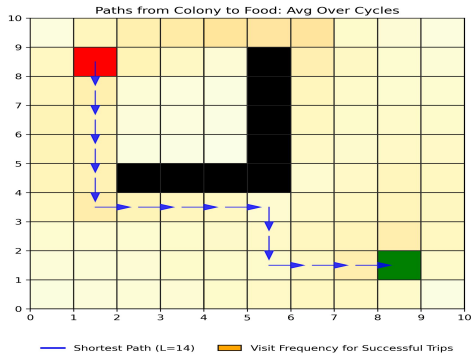
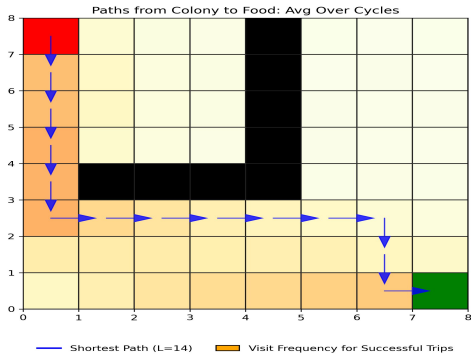
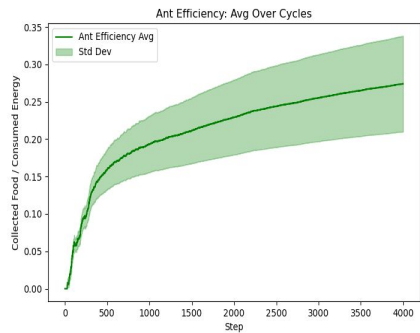
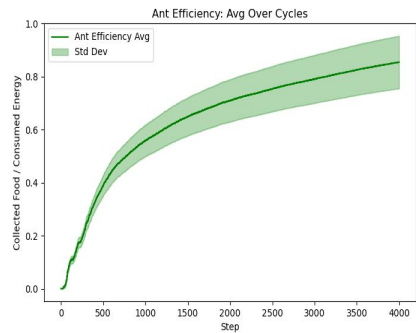
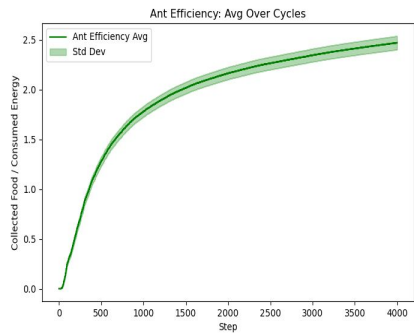
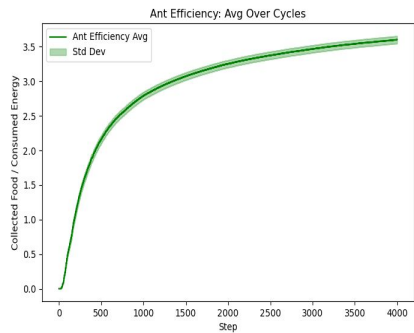


1000



5000

Environments: Size



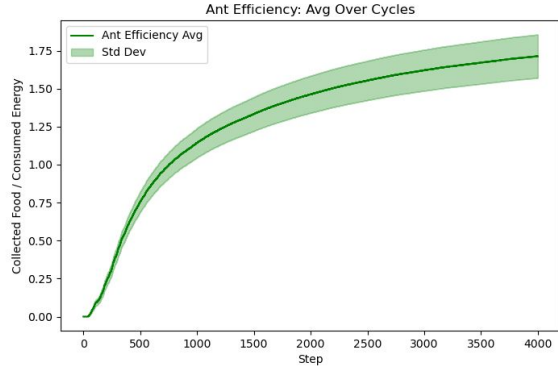
8x8

10x10

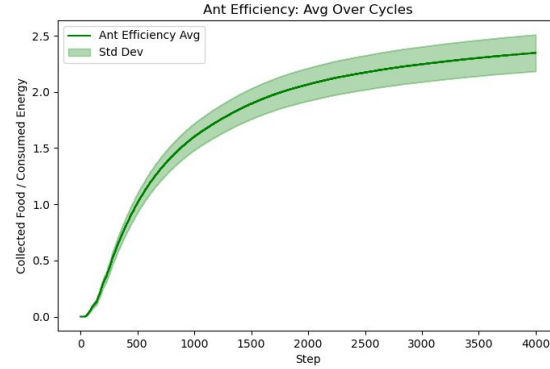
12x12

14x14

Elitist: No constraints

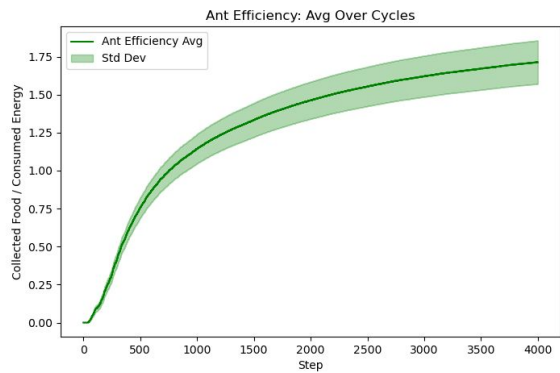


Without Elitist

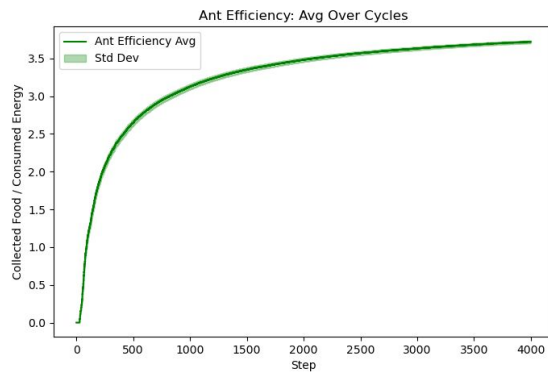


With Elitist

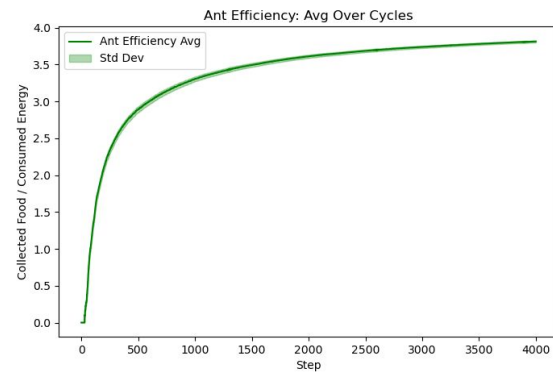
Movement Rules



No Constraints

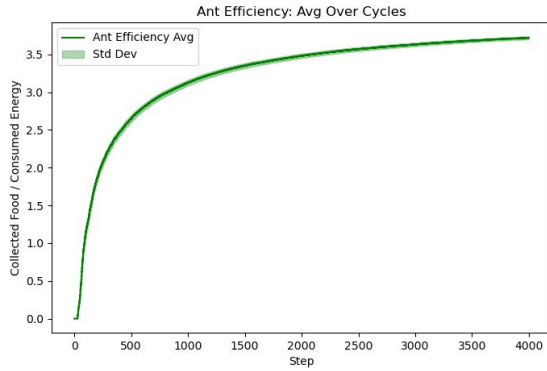


No Backtracking

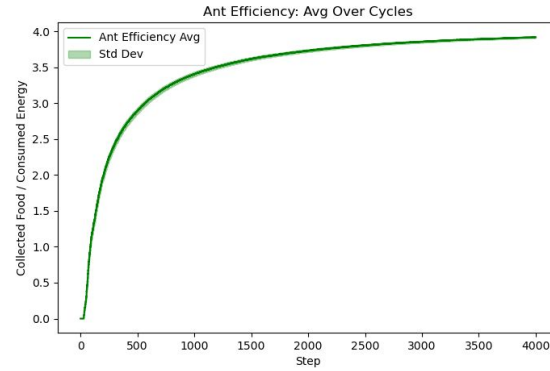


Self Avoiding

Elitist: No Backtracking

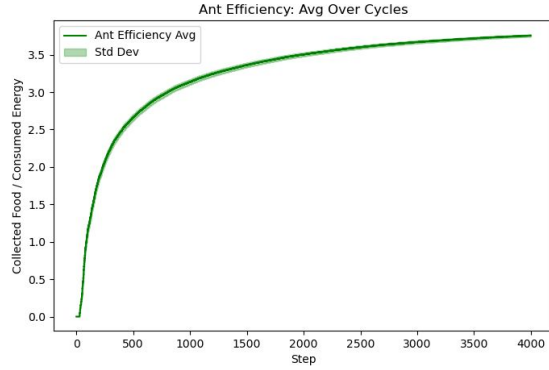


Without Elitist

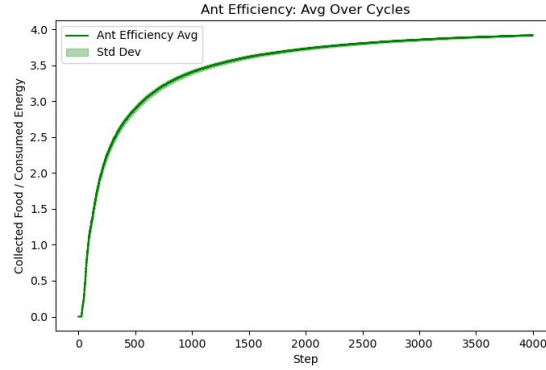


With Elitist

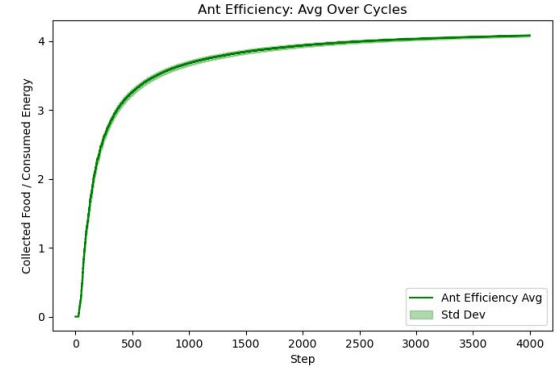
Elitist Pheromone Strength



0.001

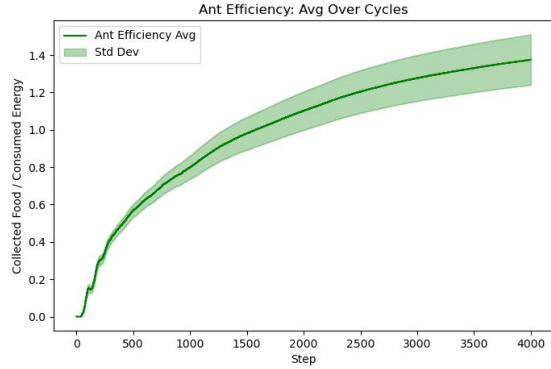


0.01

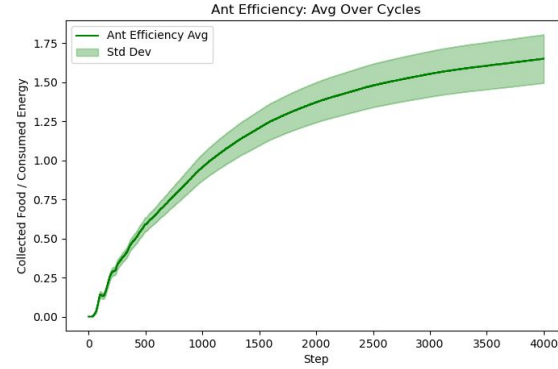


0.1

Reinforcement Model for Movement



Without Elitist



With Elitist

Code Performance and Improvements

```
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update:/Documents/stu.. print_figure:/nix/store/3bnj8jbm7d6iipzv1lib9hr0c38a7850-python3-3.13.3-env/lib/python3.13/site-packages/matplotlib/backend_bases.py:2049
_draw_frame:/nix/store.. savefig:/nix/store/3bnj8jbm7d6iipzv1lib9hr0c38a7850-python3-3.13.3-env/lib/python3.13/site-packages/matplotlib/figure.py:3334
b.. _draw_next_frame:/nix/.. grab_frame:/nix/store/3bnj8jbm7d6iipzv1lib9hr0c38a7850-python3-3.13.3-env/lib/python3.13/site-packages/matplotlib/animation.py:344
av.. save:/nix/store/3bnj8jbm7d6iipzv1lib9hr0c38a7850-python3-3.13.3-env/lib/python3.13/site-packages/matplotlib/animation.py:943
<module>:/Documents/study/y1/q3/CP/p3/main.py:1
```

```
bootstrap_error:/Documents/study/y1/q3/CP/p3/code/plots.py:260
an.. an.. average_steps_per_ant:/Documents/study/y1/q3/CP/p3/code/plots.py:228
<module>:/Documents/study/y1/q3/CP/p3/main.py:1
```

print_figu..
savefig:/n..
savefig:/n..
plot_paths_on.. p.. vi..

```
_cal.. up.. n..  
exec.. de.. n..  
_loa.. de.. n..  
_find_.. cl.. n..  
_find_.. col.. f..  
explore_grid:/Documents/study/y1/q3/CP/p3/code/ants.py..  
next_step:/Documents/study/y1/q3/CP/p3/code/ants.py:45  
<module>:/Documents/study/y1/q3/CP/p3/main.py:1
```

choices:/nix..
print_pn..
<lambda>..
print_fi..
savefig..
savefig..
plot_glo.. savefig..

Outlook

- Movable obstacles
- Pre-trained reinforcement
- Limited/replenishing food sources

Questions?

